ECON 2050 – Mathematical Economics

Basic Real Analysis IV

The simplest chain rule: A review

In the previous lecture, we discussed the following result, usually referred to as the chain rule:

Theorem

Let $D \subseteq \mathbb{R}$ be open, and let $f : D \to \mathbb{R}$ be differentiable at $x \in D$. Let $g : E \to \mathbb{R}$, where $f(D) \subseteq E$, be differentiable at $f(x)$, then the composition $g \circ f$ is differentiable at x, and

$$[(g \circ f)(x)]' = g'(f(x))f'(x).$$

This rule works nicely when we work with two functions of one variable.

However, sometimes we are interested in situations where the "outer function" is a function of several variables, and all variables are affected by the same parameter.

Example: Suppose the production function of a company is $F(K, L)$, where K and L denote capital and labor, respectively. It is natural to assume that both capital and labor accumulate over time.

In this example, denoting time by t, we see that both K and L are functions of t.

A more general chain rule

Theorem

Suppose that $f : \mathbb{R}^k \to \mathbb{R}$ *is differentiable, and for each* $j = 1, 2, \ldots, k$, $g_j : \mathbb{R}^m \to \mathbb{R}$ *is differentiable. Then the function* $F : \mathbb{R} \to \mathbb{R}$ *given by*

$$F(x) \equiv f(g_1(x), \ldots, g_k(x))$$

is differentiable, and

$$F'(x) = f_1(g_1(x), \ldots, g_k(x))g_1'(x) + \ldots + f_k(g_1(x), \ldots, g_k(x))g_k'(x),$$

where f_j *is the partial derivative of* f *with respect to its* j*-th coordinate.*

Example (cont'd): Suppose further that at a given moment in time,

- the marginal product of labor is 2.5 and the marginal product of capital is 3,
- the amount of capital is increasing by 2 each unit of time and the rate of change of labor is 0.5.

What is the rate of change of output?

The first bullet point indicates that

$$\frac{\partial F(K,L)}{\partial K} = 3, \quad \text{and} \quad \frac{\partial F(K,L)}{\partial L} = 2.5.$$

By the second bullet point, we know that $K'(t) = 2$ and $L'(t) = 0.5$. Then by the (generalized) chain rule,

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\partial F(K,L)}{\partial K} K'(t) + \frac{\partial F(K,L)}{\partial L} L'(t) = 2.5 \cdot 2 + 3 \cdot 0.5 = 6.5.$$

Exercise: Consider the production function $Q = 2K^2L$. Suppose that the inputs K and L vary with the interest rate r, via the expressions

$$K(r) = \frac{1}{\ln(r+1)}, \quad \text{and} \quad L(r) = 100 + 25r.$$

What is the rate of change of output Q?

We apply the chain rule:

$$\begin{aligned}
\frac{\mathrm{d}Q}{\mathrm{d}r} &= \frac{\partial Q}{\partial K} \cdot K'(r) + \frac{\partial Q}{\partial L} \cdot L'(r) \\
&= 4K(r)L(r) \cdot \left(-\frac{1}{(r+1)[\ln(r+1)]^2} \right) + 2K^2(r) \cdot 25 \\
&= -\frac{400 + 100r}{(r+1)[\ln(r+1)]^3} + \frac{50}{[\ln(r+1)]^2}.
\end{aligned}$$

The theorem above also suggests a natural generalization of the chain rule to the case where the "inside functions" are *also* functions of several variables.

Theorem

Suppose that $f : \mathbb{R}^k \to \mathbb{R}$ is differentiable, and for each $j = 1, 2, \ldots, k$, $g_j : \mathbb{R} \to \mathbb{R}$ is differentiable. Then the function $G : \mathbb{R}^m \to \mathbb{R}$ given by

$$F(\boldsymbol{x}) \equiv f(g_1(\boldsymbol{x}), \ldots, g_k(\boldsymbol{x}))$$

is differentiable, and for each $i = 1, \ldots, m$,

$$\frac{\partial F(\boldsymbol{x})}{\partial x_i} = f_1(g_1(\boldsymbol{x}), \ldots, g_k(\boldsymbol{x}))\frac{\partial g_1(\boldsymbol{x})}{\partial x_i} + \ldots + f_k(g_1(\boldsymbol{x}), \ldots, g_k(\boldsymbol{x}))\frac{\partial g_k(\boldsymbol{x})}{\partial x_i},$$

where x_i is the i-th coordinate of $\boldsymbol{x}$, f_j is the partial derivative of f with respect to its j-th coordinate.

Second-order derivatives

Intuitively, higher-order derivatives are the result of differentiating a function repeatedly.

Consider a differentiable function $f : D \to R$, where D is an open subset of $\mathbb{R}$. Consider the "derivative function" $x \mapsto f'(x)$; we denote it by f'.

The derivative of f' at x is called the **second-order derivative** of f at x, and is written $f''(x)$.

We may go on to define even higher order derivatives (third, fourth, etc.), but we don't need those in this course.

Example: Consider the function $f(x) = x^2$. Its (first-order) derivative is $f'(x) = 2x$, and its second-order derivative is $f''(x) = 2$.

Second-order partial derivatives

Let $D \subseteq \mathbb{R}^k$ be open. If $f : D \to \mathbb{R}$ is partially differentiable, then, if they exist, the partial derivatives of the functions

$$\frac{\partial f}{\partial x_i} : D \to \mathbb{R}, \boldsymbol{x} \mapsto \frac{\partial f(\boldsymbol{x})}{\partial x_i}, \quad \forall i = 1, \ldots, k,$$

are called the **second-order partial derivatives** of f. For $j = 1, \ldots, k$, we write

$$\frac{\partial^2 f(\boldsymbol{x}_0)}{\partial x_i \partial x_j} \quad \text{or} \quad f_{x_i x_j}(\boldsymbol{x}_0)$$

for the partial derivative $\dfrac{\partial \left(\frac{\partial f}{\partial x_i} \right)(\boldsymbol{x}_0)}{\partial x_j}$ of $\frac{\partial f}{\partial x_i}$ at $\boldsymbol{x}_0$ with respect to x_j.

A twice partially differentiable function $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto f(x, y)$ has 4 second-order partial derivatives. We write ∂x^2 for $\partial x \partial x$ etc.:

1. $\frac{\partial^2 f}{\partial x^2} = f_{xx} =$ second-order partial derivative of f with respect to x and x
2. $\frac{\partial^2 f}{\partial x \partial y} = f_{xy} =$ second-order partial derivative of f with respect to x and y
3. $\frac{\partial^2 f}{\partial y \partial x} = f_{yx} =$ second-order partial derivative of f with respect to y and x
4. $\frac{\partial^2 f}{\partial y^2} = f_{yy} =$ second-order partial derivative of f with respect to y and y

Cases 2 and 3 are called **mixed** or **cross-partial derivatives**.

Example: To obtain the second-order partial derivatives of

$$f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto e^{xy},$$

we first calculate the first-order partial derivatives

$$\frac{\partial f(x,y)}{\partial x} = ye^{xy}; \quad \frac{\partial f(x,y)}{\partial y} = xe^{xy}$$

and then differentiate these with respect to x and with respect to y:

$$\begin{aligned} \frac{\partial^2 f(x,y)}{\partial x^2} &= y^2 e^{xy}; \quad \frac{\partial^2 f(x,y)}{\partial x \partial y} = e^{xy} + yxe^{xy} = e^{xy}(1+xy) \\ \frac{\partial^2 f(x,y)}{\partial y^2} &= x^2 e^{xy}; \quad \frac{\partial^2 f(x,y)}{\partial y \partial x} = e^{xy} + yxe^{xy} = e^{xy}(1+xy) \end{aligned}$$

Observe that here, the cross partial derivatives are equal.

This is often the case in economic applications. The following theorem provides sufficient condition for the cross-partial derivatives to be equal.

Theorem (Young's Theorem)

If $D \subseteq \mathbb{R}^k$ is open and $f : D \to \mathbb{R}$ twice continuously partially differentiable, then

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}, \quad \forall x_i, x_j$$

If Young's theorem holds, the order of differentiation does not matter. (In the previous example, Young's theorem applied.)

Hessian matrix

Let $D \subseteq \mathbb{R}^k$ be open, and $f : D \to \mathbb{R}$ be twice partially differentiable.

The **Hessian matrix** $Hf(\boldsymbol{x}_0)$ of f at $\boldsymbol{x}_0$ consists of all its second-order partial derivatives of f at $\boldsymbol{x}_0$:

$$Hf(\boldsymbol{x}_0) = \begin{pmatrix} f_{x_1x_1}(\boldsymbol{x}_0) & f_{x_1x_2}(\boldsymbol{x}_0) & \dots & f_{x_1x_k}(\boldsymbol{x}_0) \\ f_{x_2x_1}(\boldsymbol{x}_0) & f_{x_2x_2}(\boldsymbol{x}_0) & \dots & f_{x_2x_k}(\boldsymbol{x}_0) \\ \dots & \dots & \dots & \dots \\ f_{x_kx_1}(\boldsymbol{x}_0) & f_{x_kx_2}(\boldsymbol{x}_0) & \dots & f_{x_kx_k}(\boldsymbol{x}_0) \end{pmatrix}$$

If Young's theorem holds (which we will assume from now on) this matrix is symmetric (that is, for all i and all j, the entry in the i-th row and j-th column equals the entry in the j-th row and the i-th column).

Example (ct'ed): The Hessian matrix of

$$f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto e^{xy},$$

at (x, y) is

$$Hf(x, y) = \begin{pmatrix} f_{xx}(x, y) & f_{xy}(x, y) \\ f_{yx}(x, y) & f_{yy}(x, y) \end{pmatrix} = \begin{pmatrix} y^2 e^{xy} & e^{xy}(1 + xy) \\ e^{xy}(1 + xy) & x^2 e^{xy} \end{pmatrix}$$

Implicit functions

Example: The level curves of the function $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2$ have the form

$$x^2 + y^2 = C. \tag{1}$$

Each of f's level curves defines a function of one variable, as we can see by rearranging (1) as:

$$y = \sqrt{C - x^2}. \tag{2}$$

The difference between expressions (1) and (2) is that

- in (1) the variable y appears in the "**implicit**" form (not separated) and
- in (2) it appears in the "**explicit**" form (separated).

However, we cannot always solve for y. For example, consider

$$y^5 - 5xy + 4x^4 = 0, \tag{3}$$

an equation describing a level curve of $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto y^5 - 5xy + 4x^4$.

Note that $(x_0, y_0) = (1, 1)$ lies on this level curve, as it solves (3).

Questions:

- Can we still describe the level curve given by (3) as a function of one variable, at least "around" the point $(x_0, y_0) = (1, 1)$?
- If so, is this function differentiable? What is its slope?

Why do we care?

Suppose a firm's profit function, which is a function of both the price of the product p and the input price c, is $\Pi(p, c)$.

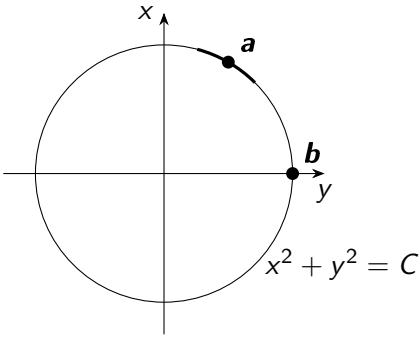
We know from ECON1050 or ECON2010 that if we wish to choose a price to maximize the profits, a necessary condition is the so-called "first-order condition"

$$\frac{\partial \Pi(p, c)}{\partial p} = 0.$$

If we wish to understand how does the optimal price p^*, which is a solution to the first-order condition, varies in c:

- for sufficiently simple profit functions we can explicitly write p^* as a function of c: if $\Pi(p, c) = 2pc - p^2/2$, the first-order condition is $2c - p^* = 0$, and therefore $p^* = 2c$
- however, for slightly more complicated profit functions, this may not be possible
- we may, nonetheless, be interested in understanding how the optimal price should change when there is a small change in the input price

Observation: In the previous example (level curves of the function $f(x, y) = x^2 + y^2$), the expression $y = \sqrt{C - x^2}$ *may not* define a function around certain points on the level curve.



Around point ***a***, y can be seen as a function of x: $y = g(x) := \sqrt{C - x^2}$; however, y doesn't define a function around point ***b***.

Theorem (Implicit function theorem)

Suppose $G(x, y)$ is a continuously differentiable real-valued function defined on an open set containing (x_0, y_0). If $G(x_0, y_0) = C$, and

$$\frac{\partial G(x_0, y_0)}{\partial y} \neq 0,$$

then there exists a $\delta > 0$ and a continuously differentiable function $g : B_\delta(x_0) \to \mathbb{R}$ such that

- $G(x, g(x)) = C$ *for all* $x \in B_\delta(x_0)$,
- $g(x_0) = y_0$, *and*
- $g'(x_0) = -\dfrac{\dfrac{\partial G(x_0, y_0)}{\partial x}}{\dfrac{\partial G(x_0, y_0)}{\partial y}}$.

Intuitively, if the assumptions of the theorem hold, even though y cannot be isolated in the original equation $G(x, y) = C$, the derivative of y with respect to x around the point (x_0, y_0) can be isolated in terms of the partial derivatives of G.

To get some intuition for the condition $\frac{\partial G(x_0,y_0)}{\partial y} \neq 0$, consider the equation (level curve) $x^2 + y^2 = 1$ and

$$G(x, y) = x^2 + y^2 = 0$$

For this G,

$$\frac{\partial G(x, y)}{\partial y} = 2y$$

and $\frac{\partial G(x,y)}{\partial y} = 0$ if and only if $y = 0$.

If $y_0 = 0$, then $x_0 = 1$, and there is no $\delta > 0$ such that we can describe the level curve by a function $g : B_\delta(x_0) \to \mathbb{R}$: take any $\varepsilon \in (0, \delta)$, *along the curve*,

- if we increase x a bit to $x = 1 + \varepsilon$, then there is no corresponding y so that $(1 + \varepsilon, y)$ solves the equation;
- if we decrease x a little to $x = 1 - \varepsilon$, then there are two equally good candidates for y near $y = 0$, namely $y_1 = \sqrt{2\varepsilon - \varepsilon^2}$ and $y_2 = -\sqrt{2\varepsilon - \varepsilon^2}$.

To understand the formula for $g'(x)$, note that by definition, if we move along a level curve of G, then the value of G does not change.

Recall that we can approximate the change in G from a change $(\mathrm{d}x, \mathrm{d}y)$ in its variables by

$$DG(x_0, g(x_0)) \cdot (\mathrm{d}x, \mathrm{d}y) = \frac{\partial G(x_0, g(x_0))}{\partial x}\mathrm{d}x + \frac{\partial G(x_0, g(x_0))}{\partial y}\mathrm{d}y$$

If we (approximately) move along a level curve, then

$$\frac{\partial G(x_0, g(x_0))}{\partial x}\mathrm{d}x + \frac{\partial G(x_0, g(x_0))}{\partial y}\mathrm{d}y = 0$$

and rearranging yields

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g'(x) = -\frac{\frac{\partial G(x_0, g(x_0))}{\partial x}}{\frac{\partial G(x_0, g(x_0))}{\partial y}}$$

Example: Consider the equation $x^2 - 3xy + y^3 = 7$ around the point $(x_0, y_0) = (4, 3)$.

Consider the level curve of level 7 of the function $G(x, y) = x^2 - 3xy + y^3$. One can show that

$$\frac{\partial G(x,y)}{\partial x} = 2x - 3y, \quad \text{and} \quad \frac{\partial G(x,y)}{\partial y} = -3x + 3y^2;$$

because both partial derivatives are continuous, G must be continuously differentiable. Moreover, $G(4, 3) = 7$ and $\frac{\partial G(4,3)}{\partial y} = 15 \neq 0$.

Then by the implicit function theorem, one can write y as a continuously differentiable function of x, denoted by $g(x)$, around $(4, 3)$, and

$$g'(4) = -\frac{\frac{\partial G(4,3)}{\partial x}}{\frac{\partial G(4,3)}{\partial y}} = \frac{1}{15}.$$

We can also write down the tangent line h to the level curve at the point $(x_0, y_0) = (4, 3)$:

$$h(x) = y_0 + g'(x_0)(x - x_0) = 3 + \frac{1}{15}(x - 4) = \frac{1}{15}x + \frac{1}{3}.$$

Exercise: Does the equation $xy\ln(xy) = 0$ define a function $y = g(x)$ around $(1, 1)$? What is its derivative at $x_0 = 1$?

Consider the level curve of level 0 of $f(x, y) = xy\ln(xy)$.

The conditions of the implicit function theorem are fulfilled with respect to f:

- $\frac{\partial f(x,y)}{\partial x} = y\ln(xy) + y$ and $\frac{\partial f(x,y)}{\partial y} = x\ln(xy) + x$, f is continuously differentiable because both partial derivatives are continuous;
- $f(1, 1) = 0$; and
- $\frac{\partial f(1,1)}{\partial y} = 1 \neq 0$.

Hence y can be written as a function of x, denoted by $g(x)$, and we can calculate its derivative at $x_0 = 1$ as

$$g'(1) = -\frac{\frac{\partial f(1,1)}{\partial x}}{\frac{\partial f(1,1)}{\partial y}} = -\frac{0+1}{0+1} = -1$$

Implicit function theorem: General case

Theorem

Suppose $\mathbf{x}_0 \in \mathbb{R}^m$, $y_0 \in \mathbb{R}$, *and* $G(\mathbf{x}, y)$ *is a continuously differentiable real-valued function defined on an open set containing* $(\mathbf{x}_0, y_0)$. *If*

$$G(\mathbf{x}_0, y_0) = C$$

and

$$\frac{\partial G(\mathbf{x}_0, y_0)}{\partial y} \neq 0,$$

then there exists a $\delta > 0$ *and a continuously differentiable function* $g : B_\delta(\mathbf{x}_0) \to \mathbb{R}$ *such that*

- $G(\mathbf{x}, g(\mathbf{x})) = C$ *for all* $\mathbf{x} \in B_\delta(\mathbf{x}_0)$,
- $g(\mathbf{x}_0) = y_0$, *and*
- $\dfrac{\partial g(\mathbf{x}_0)}{\partial x_i} = -\dfrac{\dfrac{\partial G(\mathbf{x}_0, y_0)}{\partial x_i}}{\dfrac{\partial G(\mathbf{x}_0, y_0)}{\partial y}}$ *for all* $i = 1, \ldots, m$.

Example: Consider the equation $x^3 + y^3 + z^3 - 3z = 0$.

By the implicit function theorem this equation defines a function z of x and y in a neighborhood of every point (x_0, y_0, z_0) for which A) the equation holds and B) $\frac{\partial f(x_0,y_0,z_0)}{\partial z} \neq 0$, where $f(x, y, z) = x^3 + y^3 + z^3 - 3z$.

We can calculate the partial derivatives of z by using the partial derivatives of the function f.

$$\frac{\partial z(x_0, y_0)}{\partial x} = -\frac{\frac{\partial f(x_0,y_0,z_0)}{\partial x}}{\frac{\partial f(x_0,y_0,z_0)}{\partial z}} = -\frac{3x_0^2}{3z_0^2 - 3}$$

$$\frac{\partial z(x_0, y_0)}{\partial y} = -\frac{\frac{\partial f(x_0,y_0,z_0)}{\partial y}}{\frac{\partial f(x_0,y_0,z_0)}{\partial z}} = -\frac{3y_0^2}{3z_0^2 - 3}$$